



Fruit detection and yield estimation in *Camellia oleifera* based on improved YOLOv8 and ByteTrack algorithm

Zhipeng Dong^{a,b}, Wei Long^{a,*}, Fan Yang^b, Chunlian Yu^c, Jiayi Du^b, Kailiang Wang^a, Leyan Lyu^d

^a State Key Laboratory of Tree Genetics and Breeding, Key Laboratory of Tree Breeding of Zhejiang Province, Research Institute of Subtropical Forestry, Chinese Academy of Forestry, Hangzhou 311400, Zhejiang, China

^b College of Computer and Information Engineering, Central South University of Forestry & Technology, Changsha 410004, Hunan, China

^c Changshan Country Oil Tea Industry Development Center, Changshan 324299, China

^d College of Hydraulic Engineering, Zhejiang Tongji Vocational College of Science and Technology, Hangzhou 311231, Zhejiang, China

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ABSTRACT

Camellia oleifera, an essential woody oil tree species, produces camellia oil, which is highly nutritious and edible after seed pressing. The demand for camellia oil has increased in recent years, so accurate estimates of camellia oil production are key to fruit harvesting and price setting. The rapid development of computer vision and deep learning techniques has found wide applications in agriculture, demonstrating significant fruit detection capabilities. This study proposes a surface fruit detection algorithm in *C.oleifera* based on the improved YOLOv8 model, with lightweight RepViT as the backbone of the model, a small object detection layer with p2 layer added on the multi-scale basis of FPN and PAN, and a network structure optimized by Shape IoU loss function. Meanwhile, the improved ByteTrack algorithm based on ByteTrack is utilized for multi-scale small-target tracking. Using LightGBM and linear regression to establish the relationship between the number of surface fruits and the number of individual fruits, combined with the number of typical sample trees and the average single fruit quality in the dataset, a production estimation model for the automatic detection of *C.oleifera* is constructed. The mAP value of the improved YOLOv8 algorithm on the *C.oleifera* fruit dataset reached 86.21 %, an improvement of 4.51 % compared to the original model. Combined with average single-fruit mass and the number of representative sample trees, these counts yielded whole-tree production estimates. The approach produced a fruit-number prediction R^2 of 0.945 and a yield- estimation R^2 of 0.902 across our sample trees, demonstrating strong linear relationships and validating the pipeline for automated *C. oleifera* yield forecasting.

1. Introduction

Camellia oleifera is a woody oil tree species unique to southern China, known for its high oil content (up to 90 % unsaturated fatty acids), which can promote metabolism and is considered a high-quality edible oil [1–3]. The main production areas of *C.oleifera* are located in the Yangtze River Basin, including Hunan, Hubei, Guangdong, and Guangxi [4]. The joint efforts of the Chinese government and enterprises have significantly expanded the *C.oleifera* industry in recent years. However, the alternate bearing phenomenon of economic forest tree species has led to fluctuations in the annual production of *C.oleifera*, making it difficult to stabilize the oil price of *C.oleifera*. Therefore, scientific

management and accurate production estimation are crucial [5]. The traditional method of investigating oil tea production involves hand-picking, counting, and weighing fresh fruits. Although accurate, it is time-consuming, labor-intensive, costly, and environmentally damaging. Integrating factors such as soil type, meteorological data, and farmland management for yield prediction poses challenges for data collection, integration, and quality control [6] and may not capture the complex interactions and nonlinear relationships between crop parameters [7]. In addition, the variation of the tree parameters over time and space can further affect the prediction accuracy. Therefore, a simple, fast fruit identification and counting method is urgently needed.

The object detection model in computer vision provides a new

* Corresponding author.

E-mail addresses: 17674723013@163.com (Z. Dong), yislongwei@caf.ac.cn (W. Long), yangfanf117@hnu.edu.cn (F. Yang), 540511537@qq.com (C. Yu), dujiayi@csuft.edu.cn (J. Du), wangkl163@163.com (K. Wang), 20409316@qq.com (L. Lyu).

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solution for estimating the production of *C. oleifera*. Most yield estimation studies focus on a single image but may be hampered by object occlusion and dynamic scenes. In contrast, video counting provides comprehensive target dynamics and adversarial occlusion and offers temporal information and motion features [8]. Traditional methods such as haar features, histogram of oriented gradients and adaptive boosting, support vector machine, and deformable part-based model rely on object features such as color, shape, texture, and edges [9,10] but often exhibit difficulties in natural orchard environments. Deep learning algorithms, especially those based on convolutional neural networks, have shown promise in crop detection [11]. The You Only Look Once v8 series and R-CNN series of object detection networks have demonstrated excellent performance [12–16]. For example, You Only Look Once v5 and Faster R-CNN were used for apple fruit detection, while deep simple online and realtime tracking was used for counting, with average counting errors of 20.07 % and 31.52 %, respectively [17]; ‘MangoYOLO’ is used for mango fruit detection, with an estimated error of 4.6 % to 15.2 % in orchard fruit load [18]. We trained Faster R-CNN models for orange detection and long short-term memory models for yield estimation, achieving an approximation error of 4.53 % [19].

However, tracking fruits often faces challenges such as occlusion and different lighting conditions [20,21]. To improve detection performance, Ge et al [22] introduced the YOLO DeepSORT visual object tracking network, Wu et al [23] proposed the NDMFCS automatic counting method, Farjon et al. [24] proposed the OrangeSort algorithm, Gao et al [25] selected tree trunks as single target tracking targets, and used lightweight YOLOv5 and improved DeepSORT for lotus yield counting [26]. These multi-objective tracking algorithms significantly improve the accuracy of fruit yield estimation in natural scenarios [17, 25,27,28]. Researchers have further proposed various refinements to improve the efficiency and accuracy of fruit yield estimation. Gao et al [25] proposed a YOLOv4 mini method that integrates the channel and spatial reliability discriminant correlation filter algorithm to improve the accuracy of apple yield counting. Rong et al [29] introduced an improved tomato cluster counting method that combines YOLOv5–4D and ByteTrack, effectively overcoming the challenge of automatic yield estimation in intelligent greenhouse automation. Fu [30] proposed a small YOLOv3 network with computational complexity optimization for automatically detecting kiwifruit in orchards. Xia et al [31] introduced a new method that combines the CenterNet model with a patch-matching model based on the Kuhn Munkres algorithm, successfully solving the problem of duplicate counting in continuous images of apples and oranges. Oliveira et al [32] proposed a method combining linear regression and support vector machine algorithms to solve strawberry yield estimation’s efficiency and accuracy issues. Bai et al [33] introduced a method that combines leaf area index from Landsat satellite data with the WOFOST model to improve the accuracy of yield estimation for perennial jujube crops at the field scale. Mirhaji et al [34] proposed a deep learning method based on YOLO-V4 for automatic detection and counting of oranges in orchards. Zhang et al [35] proposed a deep learning algorithm that combines OrangeYolo and OrangeSort, effectively solving the accuracy and duplicate counting problems in citrus counting.

However, most existing methods for estimating fruit yield rely on planar computation, and for counting videos, the issues of accuracy, real-time performance, and computational complexity have not been simultaneously addressed. Therefore, this study proposes a modified YOLOv8 method based on Reparameterized Vision Transformer for oil tea yield estimation in complex orchard environments. RepViT [36] combines CNN and ViT [37] to balance local and global feature extraction, making it suitable for mobile devices with limited resources. We optimize the backbone and neck of YOLOv8. We introduce state-of-the-art object tracking techniques to reduce the computational complexity of the model while increasing the accuracy and counting speed of oil palm localization.

The primary aim of this study is to improve the accuracy of fruit

detection and yield estimation for *C. oleifera* using advanced computer vision techniques. Specifically, we focus on utilizing an enhanced YOLOv8 model integrated with a multi-target tracking algorithm. The study aims to address several key objectives:

- (1) Improve the accuracy of fruit detection in *C. oleifera* using an optimized YOLOv8 model. This objective is motivated by the need to overcome challenges such as occlusion and small object detection in natural orchard environments.
- (2) Enhance the multi-target tracking performance with the improved ByteTrack algorithm. This objective is justified by the significant impact of tracking performance on fruit counting accuracy, especially in complex scenes with occlusion and varying lighting conditions.
- (3) Establish a reliable relationship between the number of detected fruits and the yield of the tree, by integrating the fruit count data with fruit quality measurements. This objective is essential for improving yield prediction accuracy, as traditional methods often fail to account for variations in fruit quality.
- (4) Develop an empirical yield estimation model for *C. oleifera* that incorporates both fruit count and fruit quality, using linear regression methods. This is necessary to address the challenge of yield estimation in varying environmental conditions, as current models often focus solely on fruit quantity without considering quality.

2. Materials and methods

2.1. Collection of phenotypic data on fruits

All image and video data used in this study were collected at the Changshan Oil Tea Experimental Base (120.825542°E, 29.748715°N) of the Research Institute of Subtropical Forestry, Chinese Academy of Forestry (Fig. 1). Three varieties of *C. oleifera* (CL4, CL40, and CL53) were selected for investigation. For each variety, twenty trees of similar age, exhibiting healthy growth and free from pests and diseases, were chosen as sampling subjects. Data collection was conducted from November 15 to 18, 2023, between 08:30 and 17:00 local time, utilizing natural lighting conditions. To ensure dataset diversity and model robustness, data were captured under various natural conditions including different lighting intensities, occlusion scenarios, and shadow variations. Two smartphone models were employed for data acquisition: (1) Xiaomi 13 (Xiaomi Inc., China): 50 MP main camera, f/1.8 aperture, 1/1.49" sensor size. (2) Apple iPhone 13 (Apple Inc., USA): 12 MP main camera, f/1.6 aperture, 1/1.88" sensor size. Both devices were set to record 4 K video at 60 fps. The data collection protocol involved circling each target tree at approximately 0.2 m/s while maintaining the smartphone at a 45° upward tilt angle. The shooting distance was maintained between 0.6–1.5 m from the tree canopy. Video data were processed by extracting frames at 30-frame intervals, resulting in adjacent images with approximately 70 % overlap. A total of 3630 images were obtained through this process. All images were annotated using Labelling, with 64,975 *C. oleifera* fruits identified and labeled using

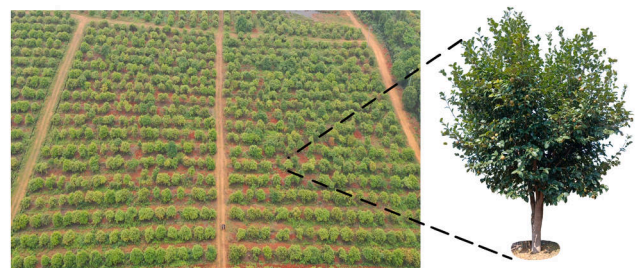


Fig. 1. Orchard layout and individual fruit tree showing spacing, fruit distribution, and conditions for yield estimation.

bounding boxes. The dataset was partitioned into training and testing subsets using an 8:2 ratio, consistent with common practices in deep learning applications [38]. The composition and treatment of the data set are shown in Table S1. During model training, data augmentation techniques including rotation, scaling, and color adjustments were implemented to enhance model generalization capability. Following image acquisition, all fruits from each tree were harvested for phenotypic characterization. Thirty fruits per tree were randomly selected for detailed measurement. Fresh fruit weight was determined using an electronic balance with 0.01 g precision. All sampling procedures were conducted in compliance with institutional guidelines and were authorized by the local forestry management authority.

2.2. Framework construction of detecting models

In this study, we employed the YOLOv8 model [39] for the detection and counting of *C. oleifera* fruits. YOLOv8 was selected due to its balanced performance in both accuracy and computational efficiency. While earlier YOLO versions, such as YOLOv4 and YOLOv5, demonstrated effectiveness in agricultural produce detection, YOLOv8 introduces several improvements. These include enhanced real-time performance, better precision in small object detection, and advanced features like the RepViT backbone and multi-scale detection layers. YOLOv8 was preferred over subsequent versions, from YOLOv9 to YOLOv12, due to its superior feature extraction capabilities and the computational efficiency of the RepViT backbone, which is optimized

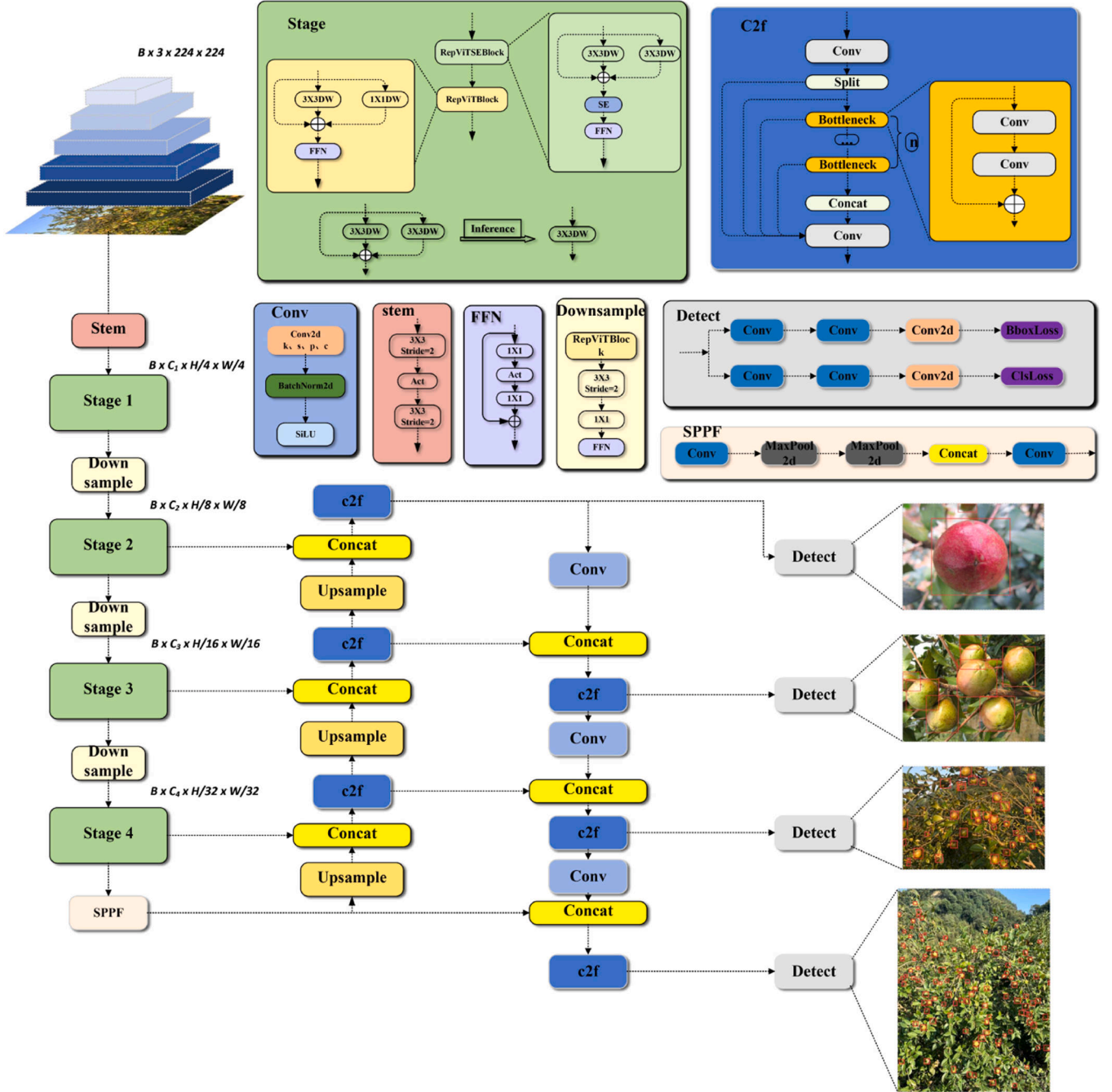


Fig. 2. Improved algorithm framework based on YOLOv8. The backbone of the improved model uses RepViT's 4 main processing stages, each with a different spatial resolution. The Stage also contains FFN (Feature Fusion Network), RepViTSEBlock, and RepViTBlock. The stages are connected by various Conv, Downsample, Upsample and feature extraction modules (C2f), which are used to process data and extract multi-scale features.

for mobile and embedded devices. These advancements allow YOLOv8 to handle the small size of *C. oleifera* fruits effectively while maintaining relatively low computational cost.

However, YOLOv8 faces challenges when detecting small *C. oleifera* fruits in complex backgrounds, particularly when targets are occluded or feature data is sparse, leading to false positives and missed detections. To address this, we enhanced YOLOv8 by improving its network structure to capture more location information. While this modification results in a slight reduction in detection speed, it significantly boosts accuracy. The improved framework consists of a backbone feature extraction network, an enhanced feature extraction network, and a prediction network (Fig. 2). The backbone employs RepViT, which combines CNN and Vision Transformer architectures, and integrates FPN and PAN networks in the Neck section. The prediction network utilizes YOLOv8's universal detection layer.

In the original YOLOv8 model, the backbone employs the CSPDarkNet-53 network. However, our improved model replaces this with RepViT, which better accommodates the constraints of mobile devices without sacrificing performance. RepViT integrates convolutional layers and ViT's self-attention mechanism after each layer, enhancing feature extraction while adapting to resource limitations through its lightweight design (Fig. 2). This structure improves performance through structural reparameterization and deep convolutions.

During the inference process, we designed the RepViTSEBlock and RepViTBlock to merge multi-branch convolutional structures into a single-branch design, reducing computational and memory overhead. Additionally, we adopted a smaller expansion ratio, uniformly set to 2, to reduce computation and channel redundancy while enhancing overall width. For input processing, RepViT uses a simplified stem module and early convolution operation, allowing it to process high-resolution input feature maps effectively and avoid information loss caused by down-sampling. Similar fine-tuned YOLOv8 approaches have been successfully applied in greenhouse environments for detecting capsicum fruits and assessing their growth stages, further confirming the adaptability of YOLOv8 for horticultural applications [40].

In the Neck section, YOLOv8 utilizes FPN and PAN networks to detect objects of various sizes. FPN captures multi-scale target features, while PAN improves small target detection by enhancing semantic feature flow (Fig. 3). The introduction of the P2 layer enables better detection of small targets, which are more challenging to detect due to their small pixel size. Related research in uncontrolled environment settings has highlighted the necessity of specialized detection and segmentation strategies when dealing with low-light or night-time scenarios, such as peduncle detection in capsicum, which underscores the importance of tailoring vision models to specific environmental

challenges [41].

Due to YOLOv8's adoption of an anchor-free approach, its loss function differs significantly from that of YOLOv5. The loss function optimization consists of two parts: classification and regression. The classification loss uses binary cross-entropy loss (BCEL), while the regression loss incorporates distribution focus loss (DFL) and bounding box regression loss (BBRL). The full loss function is expressed as shown in Eq. (1).

$$f_{\text{loss}} = \lambda_1 f_{\text{BCEL}} + \lambda_2 f_{\text{DFL}} + \lambda_3 f_{\text{BBRL}} \quad (1)$$

Unlike the CIoU loss in YOLOv8, the Shape IoU (Fig. 4) loss function is used here as the boundary box regression loss. The original boundary box regression method usually considered the geometric relationship between the GT box and the prediction box, using the relative position and shape of the boundary box to calculate the loss while ignoring the influence of inherent properties such as the shape and scale of the

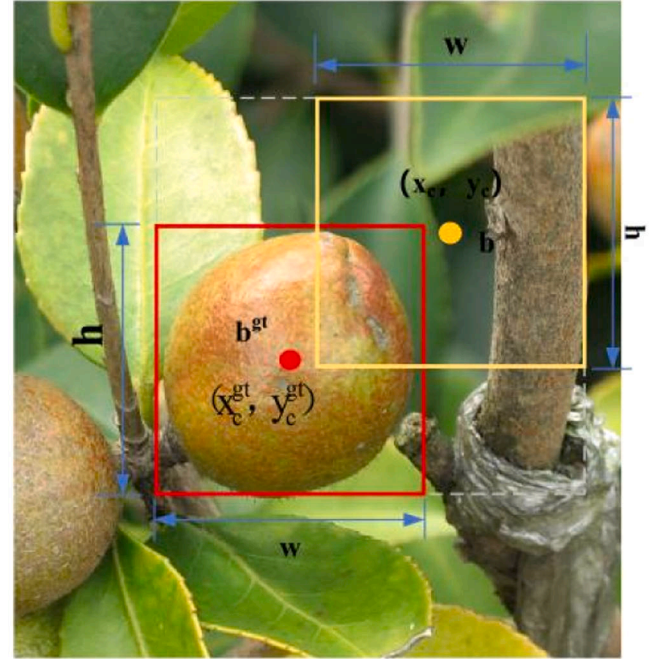


Fig. 4. Focus on boundary box shape and scale boundary box of regression methods.

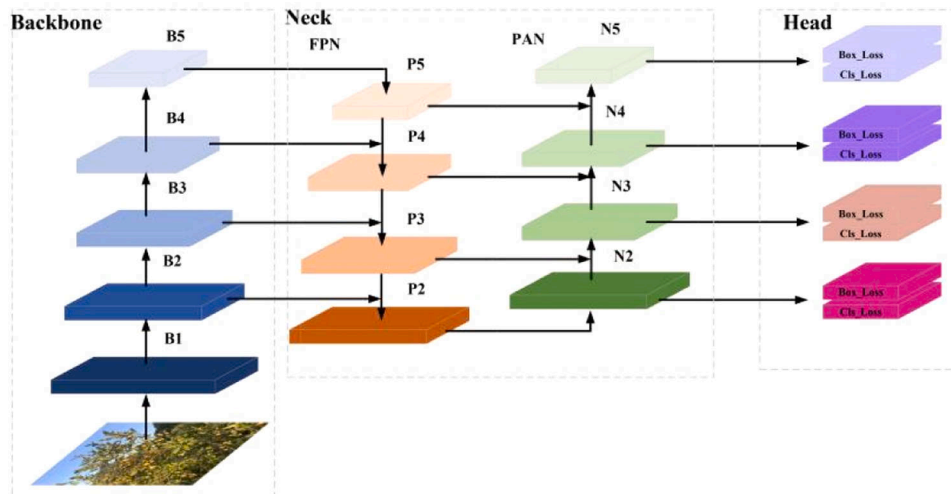


Fig. 3. Structure of improved neck.

boundary box on the boundary box regression. To remedy the shortcomings of existing studies, we use a bounding box regression method by Eq. (2)–5 that focuses on the shape and scale of the bounding box itself. This approach can make bounding box regression more accurate by focusing on the shape and scale of the bounding box itself to calculate the loss.

$$IoU = \frac{|B \cap B^{gt}|}{|B \cup B^{gt}|} \quad (2)$$

$$\omega(\omega) = \frac{2^*(\omega^{gt})^{scale}}{(\omega^{gt})^{scale} + (h^{gt})^{scale}} \quad (3)$$

$$hh = \frac{2^*(h^{gt})^{scale}}{(\omega^{gt})^{scale} + (h^{gt})^{scale}} \quad (4)$$

$$\Omega^{shape} = \sum_{t=\omega, h} (1 - e^{-\omega_t})^\theta, \theta = 4 \quad (5)$$

Here, $scale$ refers to the scaling factor related to the target size in the dataset. The variables $\omega\omega$ and hh represent weight coefficients in the horizontal and vertical directions, respectively, with their values dependent on the shape of the GT box. The corresponding bounding box regression loss is given in Eq. (6–9).

$$D = \sqrt{hh^*x_c - x_c^{gt2} + \omega\omega^*y_c - y_c^{gt2}} \quad (6)$$

$$S = \sqrt{\frac{\sum_{i=1}^M \sum_{j=1}^{N_i} \omega_{ij}^* h_{ij}}{\sum_{i=1}^M N_i}} \quad (7)$$

$$\text{Dot } D^{shape} = e^{-\frac{D}{S}} \quad (8)$$

$$L_{\text{shape-IoU}} = 1 - \text{IoU} + \text{Dot } D^{\text{shape}} + 0.5 * \Omega^{\text{shape}} \quad (9)$$

2.3. Fruit counting method based on multi-target tracking algorithm

This study uses a multi-object tracking algorithm for counting *C. oleifera* fruits. Multi-target tracking relies on computer vision to track target objects in a video sequence. The common paradigm is "Tracking by Detection" [42], which involves object detection in each frame, followed by target association to track them across frames [43]. In each frame, an object detection algorithm locates the *C. oleifera* fruits and extracts their features. A unique identifier (ID) is assigned to each target to maintain consistency and track its trajectory.

The core of the multi-target tracking algorithm is data association, which matches the predicted trajectory with the detection box in the

current frame to track the target. The accuracy of this matching determines the overall tracking precision. To improve data association, we introduce the ORB algorithm’s feature extraction and matching methods into the ByteTrack algorithm (Fig. 5). ORB, a fast and stable feature point detection algorithm, uses FAST for corner detection, and solves scale and rotation invariance with pyramid and grayscale centroid calibration. Descriptors are constructed using BRIEF, and the similarity between features is calculated.

In tracking, if a sequence exists, the Kalman filter predicts the trajectory, generating expected positions. Simultaneously, object detection is performed on the current frame. The detection confidence threshold is set to 0.001, with high (T_{high}) and low (T_{low}) score thresholds for the bounding box. High-score boxes are matched to tracked trajectories, and if matched, the Kalman filter updates the trajectory. Unmatched boxes are processed with ORB feature extraction and matching. After three consecutive frames of unmatched boxes, new trajectories are created and added to the tracking sequence.

2.4. Yield prediction method

The yield prediction of *C. oleifera* in this study was achieved through a three-stage pipeline integrating fruit detection, tracking-based counting, and regression modeling. First, an improved YOLOv8 model was employed for fruit detection, in which the RepViT backbone replaced the original CSPDarkNet structure, a P2 layer was added to the FPN-PAN neck to enhance small-object detection, and the Shape IoU loss was introduced to optimize bounding box regression, thereby enabling robust recognition of small fruits in complex orchard environments. Second, to obtain reliable fruit counts from video sequences, we adopted an improved ByteTrack algorithm following the tracking-by-detection paradigm, in which ORB feature extraction and matching were incorporated into the data association process to maintain consistent fruit identities across frames, reduce duplicate counts, and ensure accurate tracking under occlusion and variable lighting. Finally, the regression modeling process was constructed in two steps: (i) establishing a mapping between the number of detected surface fruits and the actual total fruit load per tree, where surface counts serve as the independent variable and manually measured total fruit numbers as the dependent variable; and (ii) incorporating the average weight of individual fruits, obtained from phenotypic sampling, into the model so that total yield could be expressed as the product of the predicted fruit load and the average fruit weight. Linear regression was selected as the primary modeling method due to its simplicity and interpretability, while additional machine learning regressors (e.g., Random Forest, Extra-Trees, and LightGBM) were also tested to verify robustness. Recent studies in greenhouse fruit monitoring have shown that combining multiple detection strategies, including hybrid and fusion models, can

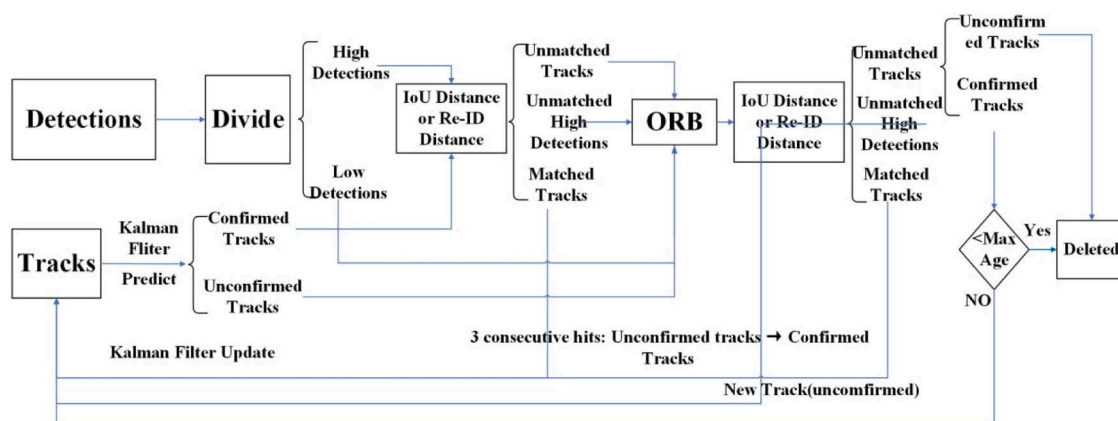


Fig. 5. Flow chart of the tracking process of the improved ByteTrack.

improve zero-shot detection and enhance generalization across different environments, which provides methodological support for incorporating multi-model regression approaches in yield prediction [44]. This integrated framework provides a systematic and reproducible method for automated yield prediction of *C. oleifera*.

2.5. Experiment evaluation metrics

In this experiment, we used the AMD EPYC 7T83 64-core processor and RTX4090 GPU for model training. The software environment includes CUDA version 11.8, Python version 3.8, and Pytorch version 2.0.0. To ensure optimal model performance during training, we adopted specific parameters and hyperparameter settings. The image size during training was set to 640×640 , with the number of iterations set to 300. To improve computational efficiency and model convergence speed, we fixed the batch size to 16 and disabled the Mosaic data augmentation technique in the last 10 epochs.

For hyperparameter selection, we employed a gradient-based SGD optimizer for model optimization. The initial learning rate was set to 0.01 and decreased to 0.0001 by the end of training. The momentum was set to 0.937, and the weight decay was set to 0.0005 to ensure stability and convergence speed during the training process. These hyperparameters were selected based on multiple experimental results and relevant literature, ensuring excellent model performance under different conditions (Table S2). This metric provides a standardized evaluation of detection accuracy under relaxed localization constraints, making it particularly suitable for real-world agricultural scenarios where precise bounding box alignment can be challenging due to small object size or partial occlusion.

To evaluate the effectiveness of the proposed method in the context of *C. oleifera* fruit detection and counting, we adopted a comprehensive set of metrics that reflect both detection accuracy and practical deployability: Precision measures the system's ability to avoid false alarms. For instance, a high precision indicates that the model rarely misclassifies leaves, branches, or shadows as fruits. In orchard automation systems, this reliability is critical for downstream applications such as robotic harvesting or yield mapping, where false positives may lead to inefficient resource allocation or inaccurate inventory estimates. Recall indicates how well the model captures all visible fruits, especially under challenging conditions such as foliage occlusion, low contrast, small target size, or uneven illumination. Since accurate yield estimation relies on minimizing missed detections, a high recall is essential for robust monitoring and effective harvest planning. mAP (mean Average Precision), particularly mAP@0.5, serves as a holistic measure of detection performance. A high mAP reflects consistent object localization and classification across diverse environmental conditions. In our application, this translates to stable performance despite variations in tree density, fruit maturity, and lighting, which are common challenges in real-world *C. oleifera* orchards. FPS (Frames Per Second) quantifies the model's inference speed, which directly affects its feasibility for real-time deployment on mobile platforms such as drones or ground robots. A higher FPS enables efficient, large-scale field monitoring without sacrificing timeliness, thus supporting practical use in precision agriculture. Together, these metrics provide a context-aware evaluation framework that goes beyond generic benchmarks, aligning technical performance with real-world agricultural requirements.

In evaluating the model's performance on the *C. oleifera* dataset, we employed various evaluation metrics, including precision, FPS, recall, and mAP. Precision (P), FPS, and recall (R) are calculated using the following Eq. (10–13):

$$P = \frac{TP}{TP + FP} \quad (10)$$

$$R = \frac{TP}{TP + FN} \quad (11)$$

$$FPS = \frac{\text{Total number of frames processed}}{\text{Total time taken(in seconds)}} \quad (12)$$

$$AP = \int_0^1 P(R) \quad (13)$$

AP represents the area under the precision-recall curve for a specific class at different confidence thresholds. mAP represents the mean average precision, which is calculated by averaging the APs for each category Eq. (14–15):

$$mAP = \frac{1}{C} \sum_{i=1}^C A \quad (14)$$

$$mAP@50\% = \frac{1}{C} \sum_{i=1}^C A P@0.5_i \quad (15)$$

Where N is the number of classes, and $A P_i$ is the average precision for class i mAP50 specifically refers to the mean average precision at an intersection over union (IoU) threshold of 0.5.

In addition, Multiple Object Tracking Accuracy (MOTA) evaluates the accuracy of multi-target tracking, mainly reflecting the accuracy of target quantity estimation and related attributes. It is used to calculate the accumulation of errors in tracking. The formula for MOTA is Eq. (16,17):

$$MOTA = 1 - \frac{\sum_i (FP + FN + IDSW)}{\sum_i g_i} \quad (16)$$

$$MOTP = \frac{\sum_{i,t} d_t^i}{\sum_i c_i} \quad (17)$$

Multiple Object Tracking Precision (MOTP) evaluates the spatial accuracy of tracked fruit positions. High MOTP ensures precise localization, which is vital for targeting or spatial yield distribution analysis. A low number of ID switches means the tracker maintains consistent identification of individual fruits across frames, which is crucial for longitudinal studies of fruit development or automated counting in dynamic scenes. In dense canopies where fruits frequently occlude one another, a high MOTA indicates that the system maintains accurate fruit identity and count over time. These metrics, interpreted within the specific context of oil tea fruit monitoring, provide a holistic assessment of the model's technical performance and its practical utility in precision agriculture.

3. Results

3.1. Comparison of training results of classical object detection models

In this study, we compared the improved YOLOv8 model with Faster R-CNN, SSD, YOLOv3, YOLOv5, YOLOv7, YOLOv8, and the newer YOLOv9 to YOLOv12 series. Faster R-CNN, a classic two-stage detector, relies on a Region Proposal Network (RPN) but suffers from high computational cost, while SSD, though faster, faces challenges in detecting small objects. The YOLO series, from YOLOv3 to YOLOv12, reformulates object detection as a regression problem, providing a good balance of speed and accuracy. Among them, YOLOv8 has been widely adopted due to its stable performance and efficiency, making it a reliable baseline model.

Our results show that YOLO models generally outperform Faster R-CNN and SSD in both accuracy and efficiency, while our improved YOLOv8 further advances performance across all key metrics (Table 1). Specifically, it achieved FLOPs, parameter count, and model size values of only 23.2 G, 6.6 M, and 14.0 MB, significantly lower than most YOLO models. At the same time, it outperformed the standard YOLOv8 in precision (85.13 % vs. 81.06 %), recall (77.84 % vs. 74.07 %), mAP50

Table 1

Performance comparison of different models.

Model	FLOPs (G)	Paras (Million)	Weight (MB)	P (%)	Recall (%)	mAP50 (%)	mAP50–95 (%)
Faster R-CNN	180	42.3	160.3	57.33	62.17	60.25	27.34
SSD	273.7	33.8	98.6	68.81	71.41	70.73	34.10
YOLOv3	193.89	61.53	120.5	73.36	66.73	68.64	32.77
YOLOv5	48.8	20.9	43.6	80.43	73.11	80.21	41.76
YOLOv7	105	37.2	74.9	79.16	74.52	79.83	40.36
YOLOv8	28.7	11.1	21.5	81.06	74.07	81.70	44.28
YOLOv9	26.9	12.0	22.8	82.10	74.00	81.11	45.12
YOLOv10	24.8	14.0	17.3	81.98	74.25	80.96	44.32
YOLOv11	22.7	15.0	18.4	81.23	74.65	81.45	44.85
YOLOv12	22.2	17.6	16.9	81.78	74.36	81.78	44.63
Improved YOLOv8	23.2	6.6	14.0	85.13	77.84	86.21	48.97

(86.21 % vs. 81.70 %), and mAP50–95 (48.97 % vs. 44.28 %).

When compared with the newer YOLOv9–YOLOv12 models, which deliver only incremental gains in certain metrics (e.g., mAP50 ranging from 80.96 % to 81.78 % and recall around 74 %), our improved YOLOv8 shows clear superiority by achieving substantially higher precision, recall, and mAP values while maintaining a smaller computational footprint. For example, despite YOLOv9–YOLOv12 requiring larger parameter sizes (12–17.6 M) and weights (16.9–22.8 MB), their performance remains lower than our improved YOLOv8, which combines lightweight design with the highest accuracy. These results highlight that the improved YOLOv8 achieves a better trade-off between efficiency and detection performance, making it more suitable for real-world applications where both accuracy and computational cost are critical.

3.2. Evaluation of the proposed detection model

In this study, the RepViT network replaces the original CSPDarkNet-53 backbone in YOLOv8. The neck design adds a P2 layer feature network, and the IoU section implements our proposed Shape IoU calculation in the loss function. To assess the effectiveness of the various improvement modules in the enhanced YOLOv8 model, ablation studies were conducted comparing YOLOv8 (baseline), YOLOv8 (RepViT), YOLOv8 (p2), and YOLOv8 (Shape IoU). These models were evaluated in terms of precision (P), recall, mAP50, and mAP50–95 (Table 2). The results show that different configurations significantly improve the object detection system, with YOLOv8 (RepViT) yielding the most notable improvements. Depending on specific requirements, selecting the appropriate configuration enhances the system's accuracy and stability.

YOLOv8 (RepViT) showed significant improvements over the baseline YOLOv8 in precision, recall, mAP50, and mAP50–95, with increases of 2.68 %, 3.36 %, 3.43 %, and 3.92 %, respectively. YOLOv8 (Neck) and YOLOv8 (Shape IoU) also showed improvements over the baseline model, with increases of up to 1.91 % in precision and recall. Ultimately, the improved YOLOv8 model outperformed all others with P, recall, mAP50, and mAP50–95 values of 85.13 %, 77.84 %, 86.21 %, and 48.97 %, respectively, surpassing the baseline YOLOv8 by 4.07 %, 3.77 %, 4.51 %, and 4.69 %.

Although box-loss, obj-loss, and cls-loss for the improved model were slightly higher than the original, precision, recall, mAP50, and mAP50–95 showed notable improvements (Fig. 6, 7). The improved YOLOv8 algorithm demonstrated excellent convergence across various

performance indicators, confirming the reliability of the *C. oleifera* fruit counting model built on this detection framework. Lower values of the three losses indicate better model training, while higher values of the four evaluation metrics indicate improved detection accuracy.

A heatmap analysis was conducted to visualize the performance improvements (Fig. 8). The first column shows the original image, the second column displays the heatmap from the baseline YOLOv8 model, and the third column shows the heatmap from the improved model. By extracting the confidence of the largest predicted value and generating the attention heatmap, we can see that the baseline model's heatmap is spread over the complex background. In contrast, the improved model focuses the heatmap on the target, with a redder color indicating higher attention in those regions. This improvement allows for more accurate object localization and identification.

To further improve interpretability, future work should focus on enhancing heatmap resolution and bounding box clarity. Clearer heatmaps and more contrasting bounding boxes will make it easier to validate detection results, especially in challenging conditions like occlusion and varying lighting.

3.3. Improved tracking method for accurate acquisition of *C.oleifera* fruits

To demonstrate the effectiveness of the improved fruit tracking method based on ByteTrack, comparative experiments were conducted with several mainstream tracking methods. The experimental results show that the improved tracking performance based on ByteTrack is better than other methods (Table 3). Perhaps because SORT, DeepSORT, and DeepMOT only retain high-confidence detection boxes for subsequent tracking, it is easy to filter out *C.oleifera* fruit detection boxes with lower confidence due to occlusion, which reduces the tracking performance of the target. ByteTrack have addressed the aforementioned shortcomings, but the algorithms overly rely on the quality of detection results, and NMS operations during the detection phase may result in filtering of available detection boxes for tracking, leading to inaccurate tracking results. Therefore, the method proposed in this study can effectively track *C.oleifera* fruits, reducing false positives and omissions. The performance comparison results of several Multi Object Tracking (MOT) methods on our *C.oleifera* dataset (Table 3) show that our method achieved the best performance in all three indicators, with MOTA of 63.7 %, MOTP of 80.2 %, and IDF1 of 77.1 %. MOTA considers the effects of false positives, false negatives, and identity switching, with

Table 2

Ablation experiment results.

Model	Backbone	Neck	IoU	P	Recall	mAP50	mAP50–95
YOLOv8 (Baseline)				81.06	74.07	81.70	44.28
YOLOv8 (RepViT)	✓			83.74(+2.68)	77.43(+3.36)	85.13(+3.43)	48.20(+3.92)
YOLOv8 (p2)		✓		82.65(+1.59)	75.98(+1.91)	83.61(+1.91)	45.93(+1.65)
YOLOv8(ShapeIoU)			✓	81.73(+0.67)	74.30(+0.27)	82.17(+0.47)	44.83(+0.55)
Improved YOLOv8	✓	✓	✓	85.13(+4.07)	77.84(+3.77)	86.21(+4.51)	48.97(+4.69)

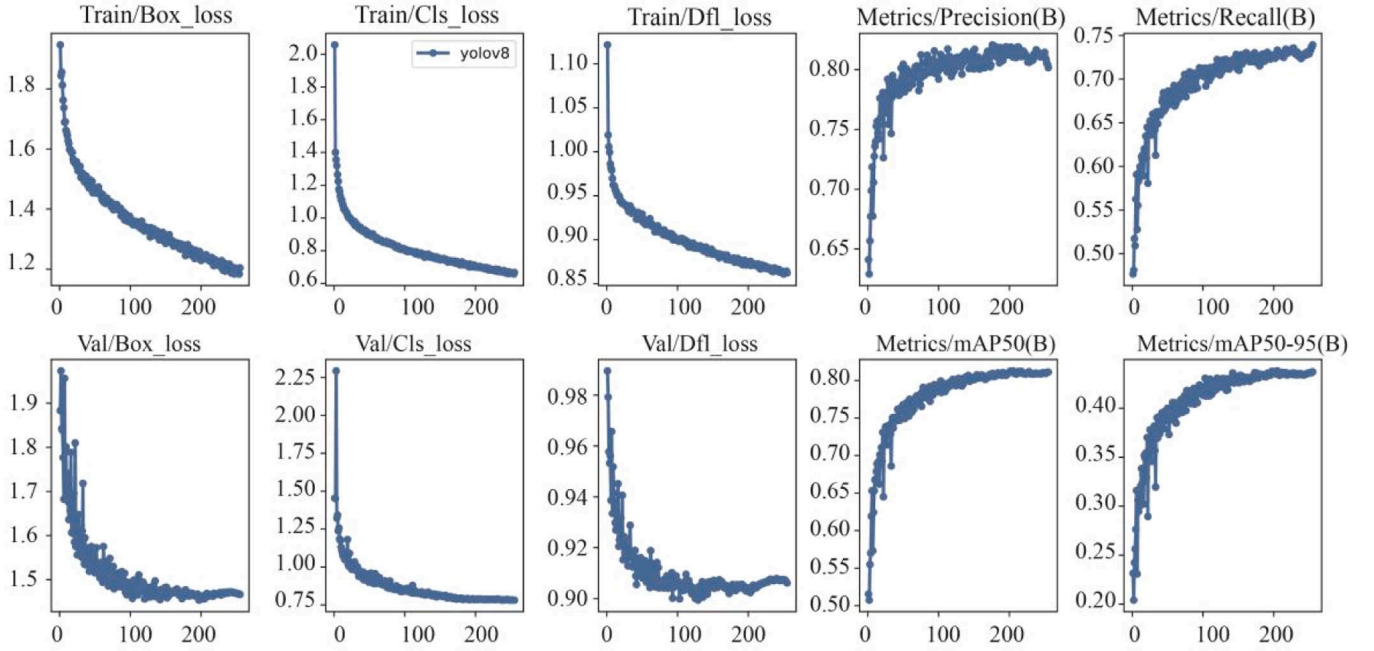


Fig. 6. The performance indicators of YOLOv8 algorithm.

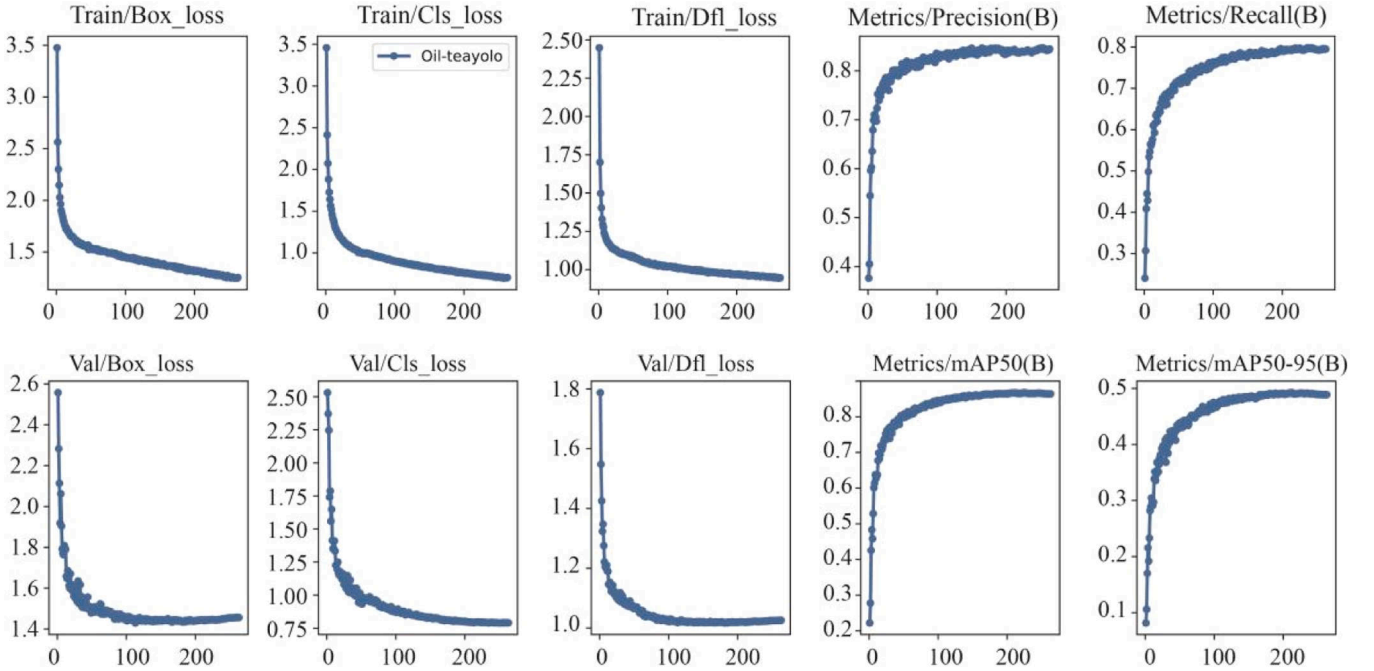


Fig. 7. The performance indicators of the improved YOLOv8 algorithm.

higher values indicating better tracking performance. MOTP measures the alignment between predicted and real bounding boxes, with higher values indicating more accurate localization. The IDF1 evaluation tracker maintains continuity with the same target, with higher values indicating better identity preservation. Overall, our method outperforms other tracking algorithms regarding accuracy, precision, and identity preservation capability.

3.4. Fruits quantity estimation of *C.oleifera*

To validate the effectiveness of the proposed counting method, we evaluated a dataset containing a large amount of counting data. To

ensure the accuracy and reliability of the evaluation, we randomly selected 60 video clips from the dataset as evaluation samples. The results showed that the counting algorithm for *C.oleifera* fruits yielded highly similar results to manual counting, indicating that our algorithm can accurately estimate the quantity of *C.oleifera* fruits. By calculating the correlation coefficient (R^2), we found a significant positive correlation between the algorithmic and manual counting results, specifically with an R^2 value of 0.945. This indicates that the algorithm can capture most trends in counting changes and has a certain degree of consistency with manual counting results.

Further analysis shows that in all test videos, the absolute counting error of most test results is less than or equal to a certain threshold. This

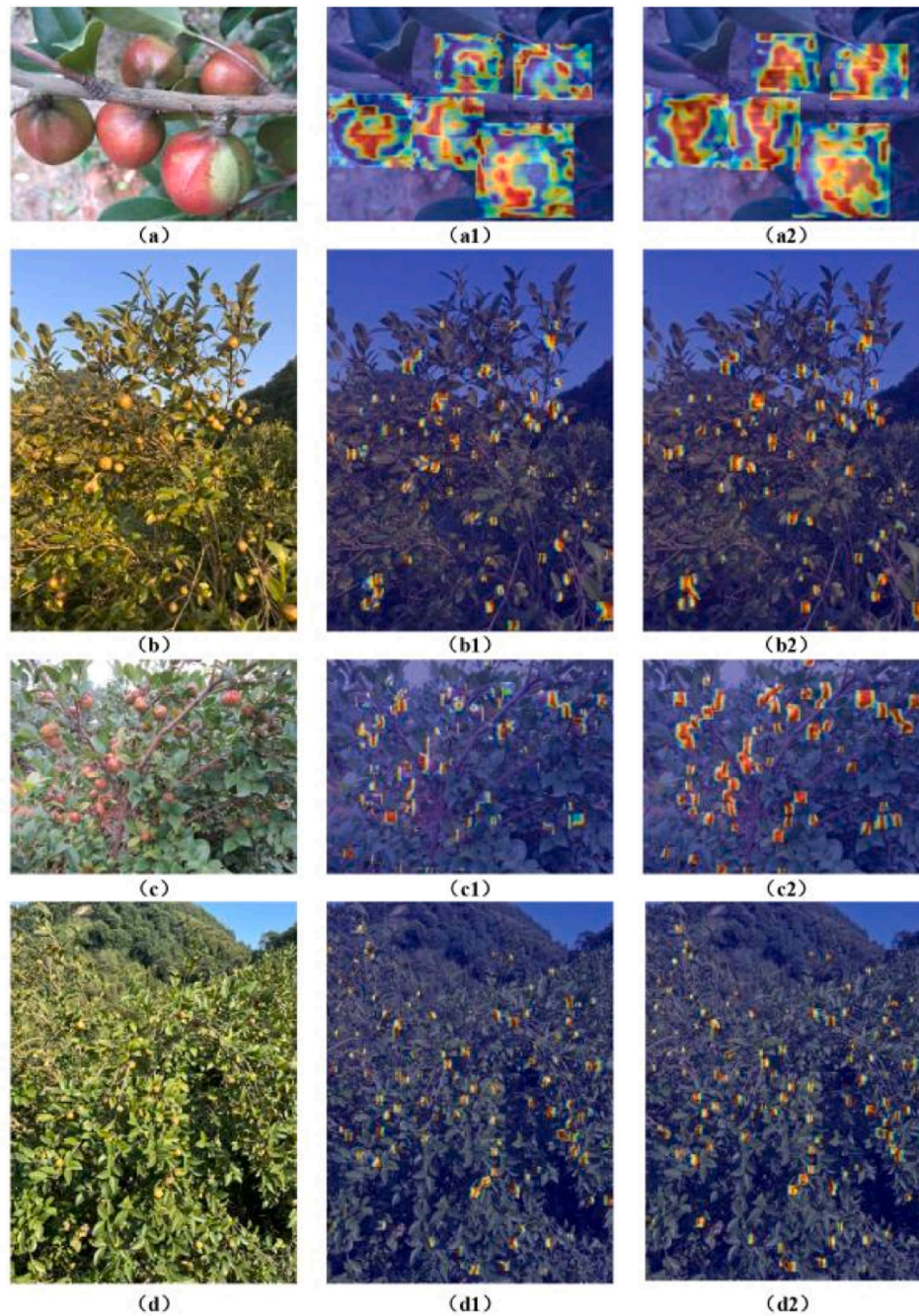


Fig. 8. Comparison of heatmaps among original image, improved YOLOv8 and YOLOv8 algorithm Original image a, b, c, d. The heatmap of YOLOv8: a1, b1, c1, d1. Improved YOLOv8: a2, b2, c2, d2.

Table 3

Experimental results of different target tracking methods on *C.oleifera* dataset.

Methods	MOTA	MOTP	IDF1
SORT	53.7	68.4	67.2
DeepSORT	59.8	73.6	70.2
DeepMOT	61.9	75.9	75.1
Bot-SORT	59.4	77.7	74.6
ByteTrack	62.6	79.4	76.7
Improved ByteTrack	63.7	80.2	77.1

means that the difference between the algorithmic and manual counting results is slight. In most cases, the algorithm can accurately estimate the number of *C.oleifera* fruits.

A strong linear correlation was observed between the actual fruit count of CL4, CL40, and CL53 and the estimated surface fruit count, with an R^2 value of 0.945, indicating a high level of consistency (Fig. 9). The scatter plots and corresponding linear regression models of the three varieties all have a fitting R^2 value exceeding 0.9, indicating that the model can predict the number of fruits well. The linear regression equation for the CL4 variety is $y = 1.1660x - 3.734$, with a fitting degree R^2 of 0.989. The linear equation for the CL40 variety is $y = 1.448x + 5.445$, with an R^2 of 0.931. Notably, the slope coefficient of 1.448 in the linear equation of the CL40 variety is mainly due to the tall tree body of the CL40 variety, which may affect data collection, resulting in a relatively large difference between actual counting and recognized counting. The linear equation for the CL53 variety is $y = 1.144x + 14.829$, with an R^2 of 0.914. The linear regression equation for

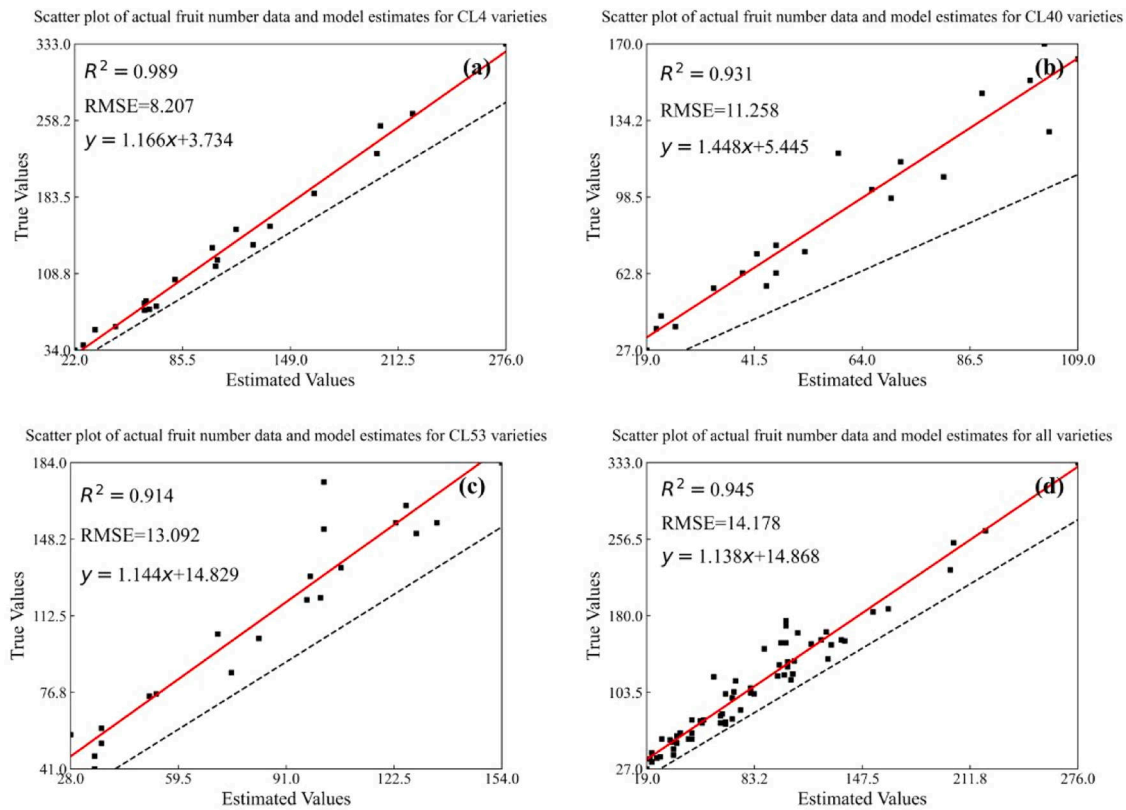


Fig. 9. Regression analysis of the actual and estimated of fruits per tree in *C.oleifera*. a:CL4. b:CL40. c:CL53. d: All varieties.

the mixed analysis of all varieties is $y = 1.138x + 14.868$, with an R^2 of up to 0.945. Compared with the previous analysis of a single variety, the fruit quantity data after mixing all varieties still maintained an excellent linear fitting effect, improving the overall R^2 value. This reflects the strong applicability and robustness of the established fruit number recognition model in cross-variety applications, and the established recognition model has good applicability and accuracy in predicting the number of fruits per plant of different varieties.

To further predict the surface fruit data of the tree, 40 sample tree data were used for model construction, and 20 reserved sample tree data were used for accuracy verification. The results showed that the existing regression models achieved good performance, with R^2 and MAPE varying between 0.622–0.901 and 9.898 % –13.859 %, respectively (Table 4). In terms of counting accuracy, the linear regression method and ExtraTrees regression method have slightly higher accuracy than other regression methods, with R^2 and MAPE of 0.901 and 0.098 and 0.901 and 0.109, respectively, indicating an excellent linear relationship between the number of surface fruits and the actual number of fruits per

plant based on *C.oleifera*. Our method has good stability and effectiveness.

3.5. Fruit yield estimation of *C.oleifera*

Various analyses were also performed during the yield prediction analysis. Good yield estimation results were achieved in all three varieties, with R^2 and RMSE ranging from 0.854 to 0.921 and 38.9 % to 55.9 %, respectively (Fig. 10). Overall shows a clear linear positive correlation between the actual fruit quantity and yield of all varieties, with a fitting coefficient R^2 of 0.854, indicating that this linear model can better describe the overall yield pattern of *C.oleifera* (Fig. 10d). The model equation $y = 0.026x + 0.26$ further quantifies this linear relation. The effect of the fit varies when subdividing into specific varieties. The CL4 has the highest fitting degree, with R^2 reaching 0.921 and a model RMSE of only 0.508, indicating good prediction accuracy (Fig. 10a). The fitting degree of the CL40 and CL53 is slightly decreased (Fig. 10b,10c). This situation may be due to significant differences in the average fruit quality per plant between varieties. To further improve the prediction accuracy, we incorporate the mean weights of individual fruit plants into the model. The specific approach introduces the number of identified fruits and the average quality of individual fruits per plant into the yield prediction model, which is fitted using linear regression.

The fitting effect of the *C.oleifera* yield estimation model was constructed based on the number of fruits and the average quality of each fruit in Fig. 11. This comprehensive analysis method significantly improves the prediction accuracy compared to previous models considering only the number of fruits. Specifically, when the critical indicator of average quality per fruit was introduced, the fitting degree of the CL4 variety further improved, with R^2 reaching 0.976 and RMSE decreasing to 0.28, indicating that the model can more accurately capture the yield pattern of the CL4 variety. The fitting degree of the CL40 variety has also improved, with R^2 reaching 0.906 and RMSE decreasing to 0.372, indicating an improvement in prediction performance. Although the

Table 4
The comparison of MSE, RMSE, MAE, MAPE, R^2 values in different regression models.

Methods	MSE	RMSE	MAE	MAPE	R^2
DT Regression	320.079	17.891	13.737	13.859	0.831
RF Regression	226.749	15.058	12.242	12.654	0.880
adaboost Regression	347.733	18.648	13.136	12.969	0.816
GBDT Regression	320.073	17.891	13.737	13.859	0.831
ExtraTrees Regression	187.741	13.702	10.836	10.989	0.901
CatBoost Regression	246.926	15.714	12.449	12.511	0.869
KNN Regression	197.798	14.064	11.305	11.545	0.895
BPnet	196.985	14.035	9.582	10.073	0.896
SVR Regression	303.184	17.412	13.314	13.286	0.840
XGBoost Regression	715.214	26.743	21.32	16.892	0.622
LightGBM Regression	187.247	13.684	9.626	9.898	0.901
Linear regression	320.079	17.891	13.737	13.859	0.831

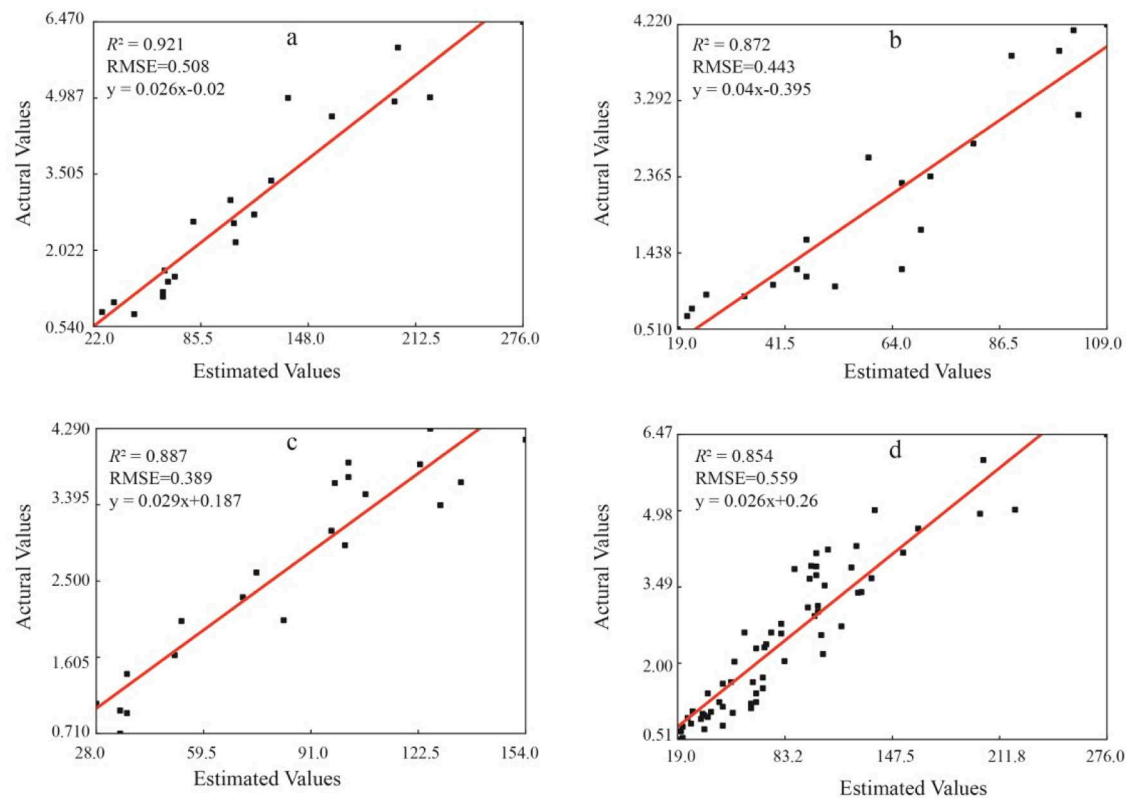


Fig. 10. Regression analysis between actual and estimated yields for different varieties of *C.oleifera*. a: CL4. b: CL40. c: CL53. d: All varieties.

fitting degree of the CL53 variety is still relatively low, it has also improved, with R^2 of 0.893 and RMSE of 0.379. By integrating data from all varieties for modeling, R^2 reached 0.905, RMSE was 0.451, and overall prediction accuracy was improved. This result suggests that the average quality of individual fruits is an essential factor in *C.oleifera* production and that including it in the model can more accurately reflect actual production patterns.

The effect of the average weight of individual fruits on the prediction of yield of *Camellia oleifera*, an essential woody oil tree species, produces camellia oil, which is highly nutritious and edible after seed pressing. The demand for camellia oil has increased in recent years, so accurate estimates of camellia oil production are key to fruit harvesting and price setting. The rapid development of computer vision and deep learning techniques has found wide applications in agriculture, demonstrating significant fruit detection capabilities. This study proposes an surface fruit detection algorithm in *C.oleifera* based on the original YOLOv8 model, with lightweight RepViT as the backbone of the model, a small object detection layer with p2 layer added on the multi-scale basis of FPN+PAN, and a network structure optimized by Shape IoU loss function. Meanwhile, the improved ByteTrack algorithm based on ByteTrack is utilized for multi-scale small-target tracking. Using machine learning methods to establish the relationship between the number of surface fruits and the number of individual fruits, combined with the number of typical sample trees and the average single fruit quality in the dataset, a production estimation model for the automatic detection of *C.oleifera* is constructed. The mAP value of the improved YOLOv8 algorithm on the *C.oleifera* fruit dataset reached 86.21 %, an improvement of 4.51 % compared to the original model. Regarding fruit number prediction, our method achieved an R^2 of 0.945. In terms of production estimation, R^2 reached 0.902. This indicates that our method has found an excellent linear fitting relationship between the number of surface fruits per plant and the measured number of fruits, and it also proves the effectiveness of our method in fitting the yield of sample trees based on the number of surface fruits. is multifaceted. It is related to the total yield of the *C.*

oleifera variety and directly affects the accuracy and applicability of the estimated model. The significant differences in average single-fruit quality among different varieties of *C.oleifera* may be due to genetic characteristics, growth environment, and nutrient absorption. If these differences are not considered in the yield estimation model, they can lead to bias in the estimation results. The direct relationship between the average weight of single fruit and yield means that even if the number of fruits is the same, the total yield of different varieties of *C.oleifera* may vary significantly. Therefore, incorporating the average weight of a single fruit into the model is a critical step in improving prediction accuracy and in the initial model, only considering the number of fruits limited the model's predictive ability. However, when we included the average quality of individual fruits in the model, not only did the fitting degree of specific varieties such as CL4 increase from 0.921 to 0.976, but the RMSE also significantly decreased from 0.389 to 0.28. In addition, the fitting degree of the CL40 variety increased from 0.854 to 0.906, and the RMSE decreased from 0.559 to 0.372. The fitting degree of the CL53 variety also increased from 0.854 to 0.893, and the RMSE decreased from 0.559 to 0.379. This proves the importance of the average weight of a single fruit in improving prediction accuracy. In addition, the sensitivity of different varieties to the average weight of a single fruit suggests that we must differentiate different varieties when establishing prediction models to ensure the model's generalization ability.

An empirical predictive model was developed based on surface fruit count data identification and the relationship between average fruit weight per plant and yield per plant. 40 sample tree data are used for model construction, and 20 reserved sample tree data are used for accuracy verification. Our method achieves an R^2 of 0.902 between the identified surface fruit counts, the average fruit weight per plant, and the individual plant yields, indicating an excellent linear fit between the surface tree fruit counts and the measured individual plant fruit counts (Table 5). This demonstrates the effectiveness of our method in fitting individual plant yields for sample trees based on crown fruit counts. decisions in resource allocation, market planning, and risk management.

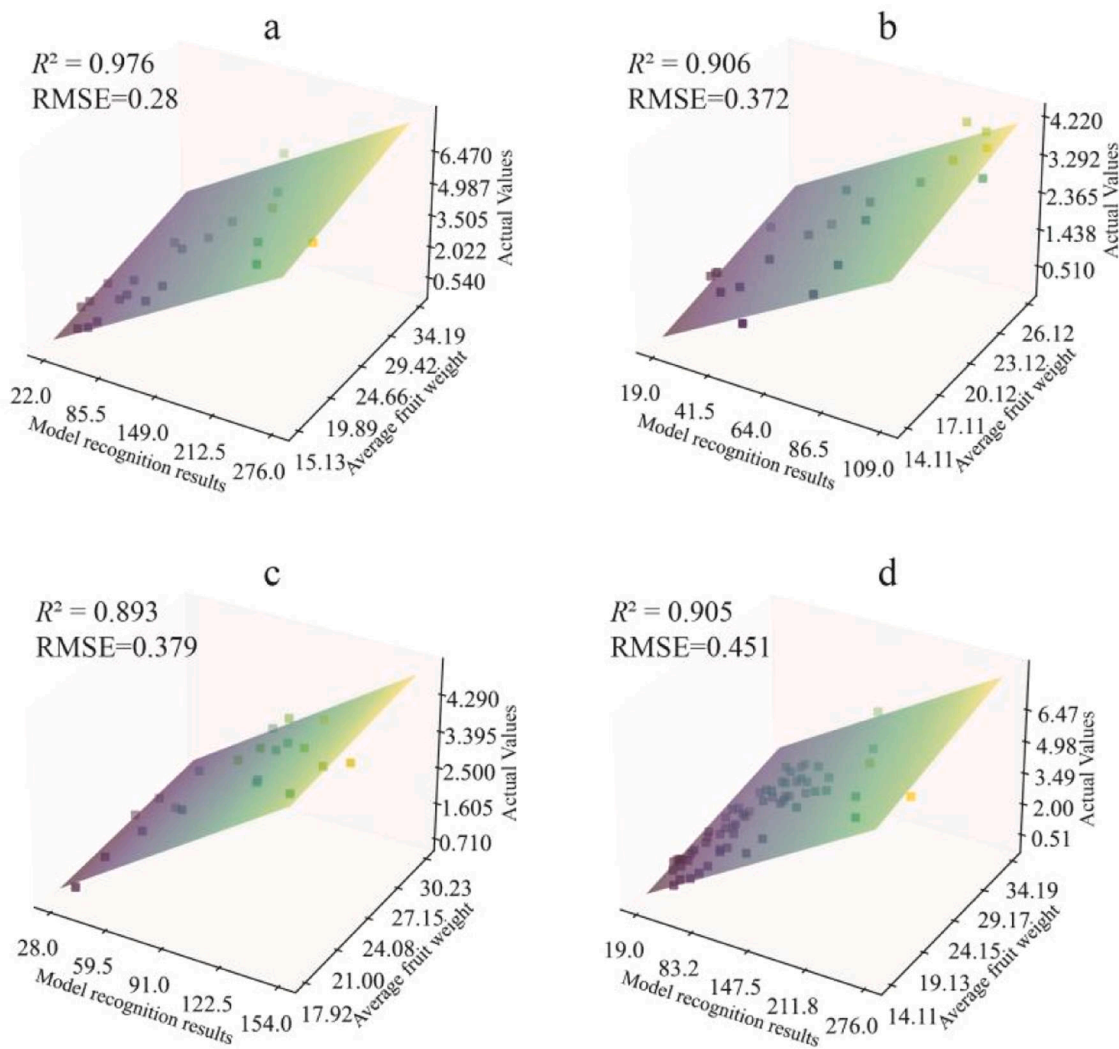


Fig. 11. Regression analysis of yield estimation for different varieties based on the number of fruits per plant and average single fruit weight a:CL4. b:CL40. c:CL53. d: all varieties.

Table 5
Regression relationship yield per plant between measured and estimation.

Methods	MSE	RMSE	MAE	MAPE	R ²
DT Regression	0.361	0.601	0.464	22.202	0.813
RF Regression	0.355	0.596	0.452	19.762	0.874
adaboost Regression	0.318	0.564	0.41	17.896	0.808
GBDT Regression	0.503	0.709	0.491	19.35	0.791
ExtraTrees Regression	0.228	0.477	0.333	16.153	0.873
CatBoost Regression	0.249	0.499	0.409	19.049	0.885
KNN Regression	0.31	0.557	0.459	17.621	0.856
BPnet	0.272	0.522	0.415	18.881	0.902
SVR Regression	0.628	0.793	0.662	119.731	0.71
XGBoost Regression	0.435	0.659	0.524	17.571	0.781
LightGBM Regression	0.537	0.733	0.573	21.303	0.743
Linear regression	0.297	0.545	0.415	17.555	0.902

Ultimately, this scientifically-based yield estimation supports the sustainable development of the oil tea industry.

4. Discussion

4.1. The improved YOLOv8 model distinctly enhances the recognition ability of *C. oleifera* fruit count

YOLOv8 has undergone various optimizations and improvements

while maintaining the high efficiency and precision features of the YOLO series. Its efficiency, accuracy, multiscale detection capability, and scalability make it a powerful tool for achieving tasks such as object detection and image segmentation. After improvement, it has been widely applied [36,45]. This study improves and integrates YOLOv8 in terms of backbone layer, Neck layer, and loss calculation, enriching the position information of feature maps without significantly reducing detection speed, thereby improving the detection accuracy of object detection algorithms. The results showed that the improved YOLOv8 model achieved an accuracy of 85.13 %, a recall of 77.84 %, and an average accuracy of 86.21 % on the *C.oleifera* dataset. Compared to the original detection model, there were improvements of 4.07 %, 3.77 %, and 4.51 %, respectively. The mAP of the improved algorithm is superior to some advanced object detection algorithms, such as the SSD algorithm, R-CNN algorithm, and other YOLO series algorithms. Our improved algorithm reduces false positives and false negatives of targets in complex environments without significantly reducing detection speed, improving detection accuracy and exhibiting superior detection performance.

Interestingly, in this study, RepViT replaced CSPDarkNet-53 in YOLOv8 and had a more advantageous enhancement effect compared to the introduction of P2 and Shape IoU, respectively. The introduction of RepViT in YOLOv8 can significantly increase the computational speed [45], show strong feature detection capabilities, and strike a good

balance between latency and accuracy. Introducing the P2 small object detection layer in the Neck layer can more sensitively detect and locate small objects, thereby improving the accuracy of small object detection. Regarding loss calculation, although existing methods consider the influence of the distance, shape, and angle between the GT box and Anchor box on bounding box regression, they ignore that the shape and size of the bounding box itself can also affect bounding box regression. In order to improve the accuracy of regression further, Shape IoU adds new geometric constraints on top of IoU to make the regression more accurate. Simultaneously, the geometric relationship between the GT box and the prediction box and the shape and size of the bounding box itself should be considered for bounding box regression. This makes the bounding box regression more accurate. From this, it can be seen that the combination of the improved YOLOv8 model and object tracking algorithm in *C.oleifera*, using the number of fruits in the video for real-time detection and recognition, can help *C.oleifera* plantation managers effectively evaluate the yield of *C.oleifera* in the forest. Solving the problem of manual production estimation takes a long time, requires a workforce, and incurs high costs. These results provide valuable insights into the tracking performance of each algorithm and help select the most suitable algorithm based on specific tracking requirements and dataset features.

4.2. The introduction of the ORB algorithm in the ByteTrack model significantly improves the accuracy and speed of multi-target tracking

Multiobject tracking tracks and localizes objects in subsequent frames based on the tracked object's position in the initial frame, associating the same object in the video. Among them, the tracking method based on the Kalman filter is widely used in the field of fruit tracking. Gan et al [46] used an extended KF-integrated autonomous navigation system to locate and track citrus fruits, but the tracking cost was expensive. To reduce costs, Wang et al [47] combined KF and Hungarian algorithms for mango tracking, reducing tracking costs and complexity, but there was a severe problem of mango occlusion. To minimize duplicate counting of fruits, Gao et al [25] proposed an improved Hungarian algorithm that preserves the tracking boxes that disappear due to occlusion during apple tracking and re-match them, improving the accuracy of apple tracking. However, for fruit trees with large or dense leaves, leaf occlusion of the fruit may result in the loss of the tracking target. To address this issue, Zhang et al [35] proposed OrangeSort based on DeepSORT, which reduces the impact of leaf occlusion on the accuracy of citrus counting results for field citrus tracking. However, this approach ignores the appearance features of the target. Ge et al [22] proposed an improved YOLODeepsort for tomato tracking, which utilizes the appearance feature extraction network in DeepSORT for association matching, meeting the practical needs of tomato yield prediction. Since the above methods are all proposed as target tracking solutions in an ensemble scenario, they are not applicable for tracking and counting *C.oleifera* in a natural scenario. The tracking algorithm proposed in this work introduces the ORB algorithm into the ByteTrack tracking model to enrich the feature information of the target. The improved tracking algorithm achieved a MOTA of 63.7 %, a MOTP of 80.2 %, and an IDF1 of 77.1 % in the *C.oleifera* fruit dataset. Compared to before the improvement, the accuracy of multi-target tracking has increased by 1.1 %, the accuracy of multi-target tracking has increased by 0.8 %, and the ID harmonic mean has increased by 0.4 %. Compared to other state-of-the-art tracking algorithms, it has a higher FPS and a more significant number of correct tracked trajectories with relatively fewer numbered jumps. When the relative distance between two targets moves closer and closer over time, traditional tracking algorithms rely solely on motion information for association matching, which may result in jumps in the number of target bins or even lead to incorrect matching or missing targets. The modified tracking algorithm performs frame-to-frame correlation matching through the dual action of motion and appearance information to improve the matching accuracy between

predicted tracking and detection boxes. It effectively improves the problem of target-number hopping and target-box loss when matching solely on a single piece of information, greatly enhancing the tracking accuracy of the tracking algorithm. So, we propose an algorithm with high accuracy and fast tracking speed.

4.3. The estimation of fruit yield based on videos provides a new idea for determining the region yield of *C.oleifera*

Yield estimation in agriculture is critical, and diverse methodologies offer their own unique merits and challenges. Statistical methods, a traditional approach, analyze historical data for yield estimation. Their simplicity suits data- or resource-constrained settings but may compromise accuracy due to oversimplifying complex growth dynamics [48]. Conversely, physical models incorporate crop growth physics and environmental factors, offering precision but necessitating intricate construction, accurate parameters, and extensive data [49]. Meteorological analysis uses weather data to predict yields [50], correlating weather with yields but possibly neglecting other influential factors. Remote sensing, especially via UAVs, facilitates large-scale yield estimation based on fruit counts. Its high spatiotemporal resolution captures crop dynamics but may be costly and less suitable for small-scale farmlands [51,52].

In contrast, video-based multi-objective tracking methods have shown unique advantages in crop yield prediction. It offers swift data acquisition, ease of use, and high accuracy. This approach utilizes refined object detection models to locate fruits precisely, assigns unique IDs via ByteTrack to prevent double counting, and employs machine learning to correlate canopy fruit counts with actual yields. This method achieved a fitting effect of 0.902 and RMSE of 0.415 in this study, demonstrating remarkable accuracy and practicality in predicting oil tea production. By integrating advanced detection and tracking technologies, this approach offers a robust solution for large-scale yield estimation, fostering the sustainable growth of the *C.oleifera* industry.

Meanwhile, our model also has inconsistent prediction yield accuracy rates among different varieties. This discrepancy can be attributed to several factors. First, dataset imbalance likely impacted the model's ability to learn from CL40, which may have been underrepresented in the training data, leading to a stronger model bias toward CL4. To address this, future work should focus on balancing the dataset or using data augmentation techniques to enhance the model's ability to generalize across different varieties. Second, the morphological similarities between CL4 and CL40, such as fruit size and shape, could contribute to difficulties in distinguishing between the two, highlighting the need for more distinct feature extraction methods. Additionally, the error distribution between predicted and actual yield values, particularly for CL40, suggests the model may be prone to false negatives, likely due to occlusion or complex background conditions, which could lead to missed detections. Further research could focus on improving the model's sensitivity to such factors, including occlusion handling and multi-view tracking.

4.4. Future applications and deployment in real-world environments

While this study focuses on improving the YOLOv8 model for fruit detection and tracking in controlled environments, future work could explore the deployment of this system in real farm or forest environments on edge devices. The successful integration of the improved YOLOv8 and ByteTrack algorithm provides a strong foundation for real-time yield estimation in precision agriculture. Deploying this method in outdoor environments would require suitable hardware, such as mobile robotic platforms or drones equipped with edge devices like Raspberry Pi or NVIDIA Jetson, which can support real-time processing while maintaining high performance. These devices, coupled with high-resolution cameras (e.g., Sony IMX series for RGB cameras or depth cameras for improved occlusion handling), could be used for fruit

detection and counting in dynamic, natural settings.

Additionally, the operational settings of these devices would need to accommodate the complexities of real-world farm environments, including varying lighting conditions, terrain, and plant structure. For instance, during fruit detection, challenges such as occlusion, leaf interference, and tree canopy variations could influence the model's performance. By deploying this method on edge devices, the model could provide real-time feedback to farmers or agricultural managers, facilitating quicker decision-making for fruit harvesting and yield estimation.

This future integration with mobile platforms would not only enhance the scalability of the method but also improve its practical applicability, enabling autonomous robots to navigate orchards or forests, detect fruits, and provide accurate yield predictions without human intervention. Future work will focus on addressing scalability issues, ensuring robustness under different environmental conditions, and exploring potential sensor fusion techniques to combine visual data with other sensory inputs for even more precise predictions.

5. Conclusion

A video sequence-based fruit counting algorithm for *C. oleifera* is proposed to address the small target characteristics of wild oil tea tree fruits and the challenges posed by different occlusion states in video sequences. The algorithm incorporates two sub-algorithms: improved YOLOv8 for fruit detection and enhanced ByteTrack for multi-object tracking. The improved YOLOv8 model, featuring a lightweight RepViT backbone, a minimal object detection layer with the P2-layer added on top of the multi-scale FPN+PAN, and a network optimized by the Shape IoU loss function, achieved an improved accuracy of 85.13 %, with mAP50 reaching 86.21 %. Additionally, the ByteTrack model, enhanced with the ORB algorithm, performed significantly better than other tracking models, showing improvements in MOTA, MOTP, and IDF1 by 1.1 %, 0.8 %, and 0.4 %, respectively.

The yield prediction model developed in this study demonstrated strong performance, with R^2 values of 0.989, 0.931, 0.914, and 0.945 for CL4, CL40, CL53, and mixed samples, respectively. The R^2 and RMSE for model accuracy tests were 0.901 and 10.989, respectively, showing good consistency between predicted and measured yield values. By incorporating the number of detected fruits and the average fruit quality, the model significantly improved production estimates, with R^2 , MSE, RMSE, MAE, and MAPE values of 0.902, 0.297, 0.545, 0.415, and 17.55, respectively.

While the proposed method has shown effectiveness in yield estimation for *C. oleifera*, some limitations must be acknowledged. The model's performance varies across different varieties, and the detection accuracy may be influenced by factors such as fruit size differences and tree size variability. Additionally, generalizing the model to other fruit types may pose challenges due to morphological differences. Environmental constraints, such as lighting conditions and occlusion, could also impact the model's robustness in diverse natural settings. Finally, scalability issues may arise when applying the model to larger or more complex orchard environments.

In future work, we plan to extend the framework to incorporate multiple-view tracking data to further improve automatic counting and yield prediction. Addressing these limitations will help enhance the model's applicability to a wider range of fruit types and real-world agricultural conditions.

Availability of Supporting Data

The datasets used and/or analyzed in the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Zhipeng Dong: Writing – review & editing, Writing – original draft, Conceptualization. **Wei Long:** Writing – review & editing, Validation,

Supervision, Funding acquisition. **Fan Yang:** Writing – original draft. **Chunlian Yu:** Validation, Supervision, Conceptualization. **Jiayi Du:** Writing – original draft, Conceptualization. **Kailiang Wang:** Writing – review & editing, Validation, Supervision. **Leyan Lyu:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.atech.2025.101435](https://doi.org/10.1016/j.atech.2025.101435).

Data availability

Data will be made available on request.

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